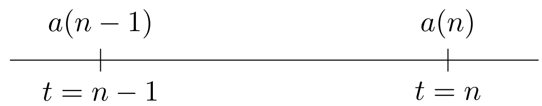
Problem 4.14

Simple interest of $i = 4\%$ is being credited to a fund. The accumulated value at $t = n - 1$ is $a(n-1)$. The accumulated value at $t = n$ is $a(n) = 1 + 0.04n$. Find n so that the accumulated value of investing $a(n-1)$ for one period with an effective interest rate of 2.5% is the same as $a(n)$.

Solution



Under simple interest,

$$a(t) = 1 + it.$$

Since $i = 0.04$,

$$a(n) = 1 + 0.04n.$$

Then

$$a(n-1) = 1 + 0.04(n-1) = 1 + 0.04n - 0.04 = 0.96 + 0.04n.$$

The effective interest rate for the period from $n-1$ to n is

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)}.$$

We are told this effective rate is $2.5\% = 0.025$, so

$$\frac{a(n) - a(n-1)}{a(n-1)} = 0.025.$$

Substituting the expressions for $a(n)$ and $a(n-1)$ gives

$$\frac{(1 + 0.04n) - (0.96 + 0.04n)}{0.96 + 0.04n} = 0.025.$$

Simplifying the numerator,

$$\frac{0.04}{0.96 + 0.04n} = 0.025.$$

Hence,

$$0.04 = 0.025(0.96 + 0.04n) = 0.024 + 0.001n.$$

Therefore,

$$0.04 - 0.024 = 0.001n,$$

so

$$0.016 = 0.001n,$$

and thus

$$\boxed{n = 16}.$$